

SUMAN BANERJEE

Machine Learning & AI Developer

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PROFESSIONAL SUMMARY

Machine Learning & AI Developer with hands-on experience building deep learning, classification, and Generative AI systems in Python. Skilled in model development and evaluation (TensorFlow/Keras, Scikit-learn, XGBoost), time-series forecasting (LSTM/GRU), and applied Generative AI (RAG, Prompt Engineering, multimodal LLM integration). Experienced in handling large, imbalanced real-world datasets end-to-end, from preprocessing through model evaluation on metrics like AUC and Recall.

CORE SKILLS

Machine Learning & Deep Learning: Deep Learning (LSTM, GRU), Classification (XGBoost), Model Evaluation (AUC, Precision/Recall), Imbalanced Data Handling (SMOTE), Time-Series Forecasting, Predictive Modeling

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Multimodal AI (Google Gemini Vision API)

Programming & Frameworks: Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Flask/FastAPI, Streamlit, SQL, Git/GitHub

Additional: Statistical Modeling, Exploratory Data Analysis (EDA), Power BI, DAX

TECHNICAL PROJECTS

Machine Learning & Deep Learning

- Fraud Detection System:** Built an XGBoost classification model to detect fraudulent activity in a highly imbalanced financial dataset of 6.3M+ transactions. Applied SMOTE for class imbalance, achieving 99.98% AUC and 98.7% Recall, minimizing false negatives.
- Hybrid Sales Forecasting Model:** Engineered a two-stage deep learning architecture combining an LSTM network for macro-trend forecasting with a GRU model for high-frequency residual error correction, processing 10 years of financial time-series data.
- Customer Lifetime Value (CLV) Prediction:** Built a Gradient Boosting model on 5,878 customers (\$17.74M revenue) using RFM feature engineering, reaching 97.45% accuracy across 7 customer segments.

Generative AI & Applied LLM Systems

- Veda AI (Automated Grading System):** Developed a multimodal Generative AI application using the Google Gemini Vision API to read and grade handwritten answer sheets — mapping handwritten answers to extracted questions and overlaying bounding boxes to cut manual grading time.
- Briefly — AI Meeting Muse:** Built a Chrome extension using Google Gemini and FastAPI to generate AI-powered meeting summaries.

Data Analytics & BI (Supplementary)

- Built Power BI dashboards for credit card usage/default-rate analytics, e-commerce sales trends, HR attrition/retention, and Paralympics performance data using advanced DAX calculations.

EXPERIENCE & INTERNSHIPS

Machine Learning Intern — Prodigy (Remote)	Aug 2025
Data Analyst Intern — Codec (Remote)	Nov 2025 – Dec 2025
Data Analyst Intern — NextGen Technohub (Remote)	Mar 2026 – Jun 2026
Data Analytics Intern — Unified Mentor (Remote)	Mar 2026 – Jun 2026
Data Analytics Intern — Zeetheta (Remote)	Aug 2026

EDUCATION & CERTIFICATIONS

B.Tech, Computer Science and Engineering — University of Kalyani | CGPA: 7.23/10 | 2023–2026

Diploma in Computer Science & Technology — State Council | CGPA: 8.0/10 | 2020–2023

IBM Data Analyst Professional Certificate — Coursera/IBM (Ongoing)

Introduction to Data Analytics — Coursera (Jul 2026)

IBM Excel for Data Analysis — Coursera/IBM (Jul 2026)